Student Oral Presentation Guidelines

1. Objectives

The purpose of this activity is to explore how IoT concepts are used to solve real-world problems. Your group will explain the assigned technical topic through a specific application, supported by clear diagrams, technical reasoning, and a discussion of challenges.

2. Topic allocation

Some technical topics are assigned to more than one group, but each group has a different application. Do not submit identical presentations: explain how the assigned technology or method fits your specific scenario.

Gr	Technical topic	Assigned application	Chapter
1	MQTT vs CoAP: protocol comparison	Smart agriculture and soil-moisture monitoring	Chapter 2
2	MQTT vs CoAP: protocol comparison	Remote patient monitoring in healthcare	Chapter 2
3	MQTT vs CoAP: protocol comparison	Smart-home lighting and energy control	Chapter 2
4	IPv6 and 6LoWPAN in constrained IoT networks	Smart-city environmental sensors	Chapter 2
5	IPv6 and 6LoWPAN in constrained IoT networks	Forest-fire detection using wireless sensors	Chapter 2
6	IPv6 and 6LoWPAN in constrained IoT networks	Precision-agriculture sensor networks	Chapter 2
7	Cloud-based vs fog-based IoT architecture	Connected vehicles and road safety	Chapter 2
8	Cloud-based vs fog-based IoT architecture	Smart traffic-light management	Chapter 2
9	Cloud-based vs fog-based IoT architecture	Industrial production-line monitoring	Chapter 2
10	IoT system design methodology	Smart irrigation system	Chapter 3
11	IoT system design methodology	Smart waste collection	Chapter 3
12	Designing an end-to-end IoT system	Smart-home security and intrusion detection	Chapter 3
13	Designing an end-to-end IoT system	Cold-chain monitoring for vaccines or food	Chapter 3
14	Developing an IoT solution using Python and REST APIs	Smart electricity-meter monitoring	Chapter 4
15	Developing an IoT solution using Python and REST APIs	Air-quality monitoring dashboard	Chapter 4
16	From IoT device simulation to cloud application	Smart greenhouse using simulated sensors and a cloud dashboard	Chapters 3–4
17	From IoT device simulation to cloud application	Water-quality monitoring with simulated sensors and a cloud dashboard	Chapters 3–4

3. Required report, presentation slides, and ZIP submission

Submit one ZIP file containing the written report and the PowerPoint file. Name the ZIP file using this format: *Example: Group_05_IPv6_and_6LoWPAN_in_Smart_Cities.zip*. Each group must prepare a written report of at least 20 pages. The report should explain the assigned technical topic and application in sufficient detail, using diagrams, technical analysis, and reliable references.